

# SARAH LIBANORE

Postdoctoral Fellow

## Keywords:

Observational Cosmology  
Large-scale structure of the Universe  
Intergalactic Medium

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## RESEARCH INTERESTS

Postdoctoral researcher in astrophysics and cosmology, focused on new observables and cross-correlations. Currently working on line-intensity mapping to study the epoch of reionization and cosmic dawn, with past experience in gravitational waves and large-scale structure tracers. Passionate about teaching, mentoring students, collaborative research, and outreach.

## EDUCATION

University of Padova  
09.2018 - 05.2022

### Ph.D. in Physics

Advisors: Prof. Michele Liguori, Prof. Alvise Raccanelli  
Thesis: *The importance of clustering analysis in future GW surveys*

University of Padova  
09.2016 - 09.2018

### Master degree in Astronomy (110/110 cum laude)

Advisors: Prof. Michele Liguori, Prof. Nicola Bartolo  
Thesis: *Gravitational Wave detections: implications for Cosmology*

University of Padova  
09.2013 - 09.2016

### Bachelor degree in Astronomy (110/110 cum laude)

Advisor: Prof. Sabino Matarrese

## PROFESSIONAL EXPERIENCE

DAMPT, U.Cambridge  
Starting 10.26

### Research Associate in Cosmology and Gravitation

Advisor: Prof. Paul Shellard, James Fergusson

Ben Gurion University  
01.2023 - 09.2026

### Azrieli International Postdoctoral Fellow

Advisor: Prof. Ely D. Kovetz  
Research Experience: line intensity mapping, simulation and forecast of 21-cm and star forming lines in the context of current surveys

University of Padova  
03.2022 - 12.2022

### Assegno di Ricerca Pre-Dottorato

Advisor: Prof. Michele Liguori

## GRANTS & AWARDS

- **BGU/Philadelphia academic bridge grant** (2025, co-PI with Profs. A. Lidz, E. Kovetz)
- **Balzan award**, held by Center for Cosmological Studies (2023, research visiting to JHU)
- **Azrieli International Postdoctoral Fellowship program** (2022)
- **Fondazione Aldo Gini scholarship** (2022)
- **PhD Sandwich Scholarship**, held by Council for Higher Education in Israel (2022)
- **Mille e una Lode**, held by University of Padova (2016)

## REFERENCES

- **Ely D. Kovetz**  
Ben Gurion University  
kovetz@bgu.ac.il
- **Michele Liguori**  
Università degli studi di Padova  
michele.liguori@unipd.it
- **Julian B. Muñoz**  
University of Texas, Austin  
julianbmuno@utexas.edu
- **Alvise Raccanelli**  
Università degli studi di Padova  
alvise.raccanelli.1@unipd.it

## TEACHING & MENTORING

- **Tutorial** on “21cmFirstCLASS” (Santander, Spain, February 2025)
- **Assistant Lecturer** for “Cosmic Tensions” Ph.D. course (GGI school, Italy, March 2023)
- **Mentor of Ph.D. and Master students in their research projects**

### CURRENT

H. Lazare (PhD, joint 21-cm+UVLF constraints)  
K. Finish (PhD, fnl effect on 21-cm at EoR)  
T. Kadosh (MsC, island model of EoR)

### PAST (5 papers)

T. Adi (PhD, primordial magnetic fields and LIM)  
J. Flitter (PhD, 21-cm during the EoR)  
G. Shmueli (MsC, neutrino masses and LIM)  
M. Peron (MsC, bias of GW mergers)

## COLLABORATIONS & MEMBERSHIPS

- Square Kilometer Array Observatory (SKAO)
- International Astronomical Union (IAU)
- European Consortium for Astroparticle Theory (EUCAPT)
- ULTRASAT
- COMAP

## OUTREACH & DISSEMINATION

PLaNCK! magazine  
2016 - ongoing

### Editorial Board of science magazine for children

Article writing and editing  
Collaboration in public relations and social media management

APS Accatagliato  
2016 - ongoing

### Outreach activities tailored to diverse audiences

Planning, design, and implementation of workshops, festivals, and events  
Science trainer & educator in schools, workshops, professional internships  
Back-office support activities

Professione... Scienza  
2019-2021

### Educational Game

I collaborated in the development of educational game cards designed to engage young people with the work and role of scientists in scientific and societal progress. The cards were created with a focus on diversity and inclusion across gender, ethnicity, disability.

Penne di Scienza  
2024 - ongoing

### Correspondence with school classes

Engaged with teenagers via letters and online meetings to foster science communication and provide role modeling

## LANGUAGES



## LIST OF PUBLICATIONS

### UNDER REVIEW

- E.D. Kovetz, H. Lazare, **S. Libanore**, J.B. Muñoz, E. Vanzan (2026)  
**arXiv:2605.13967 (submitted to APJ Lett.)** - *When galaxies burst: enhanced shot-noise for line-intensity mapping in the JWST era*
- E. Th  lie, **S. Libanore**, Y. Sklasnky, J.B. Mu  oz, E.D. Kovetz (2026)  
**arXiv:2602.12277 (submitted to Phys. Rev. Lett.)** - *Reionization Bubbles from Real-Space Cross Correlations of Line Intensity Maps*

### PUBLISHED

- 18. S. Libanore**, E.D. Kovetz, J.B. Mu  oz, Y. Sklasnky, E. Th  lie (2026)  
**arXiv:2509.08886v2 (accepted on Phys. Rev. D.)** - *A New Boundary Condition on Reionization*
- 17. M. Bosi, S. Libanore**, N. Bellomo, et al. (2026)  
**(accepted for publication)** - *Chapter in Advancing Astrophysics with the SKA – II: Using SKAO to understand the clustering of gravitational wave sources*
- 16. D. Sarkar et al. (including S. Libanore) (2026)**  
**(accepted for publication)** - *Chapter in Advancing Astrophysics with the SKA – II: Cosmology with Multi-Wavelength Line Intensity Mapping Synergies in the SKAO Era*
- 15. S. Majumdar et al. (including S. Libanore) (2026)**  
**(accepted for publication)** - *Chapter in Advancing Astrophysics with the SKA – II: Cosmology with Intensity Mapping via statistics beyond the power spectrum in the SKAO Era*
- 14. S. Libanore**, J.B. Mu  oz, E.D. Kovetz (2025)  
**Phys. Rev. D 112, 083552 (2025), arXiv 2507.15922** - *Effective model for line intensity mapping: Auto- and cross-power spectra in the cosmic dawn and reionization*
- 13. S. Libanore**, S. Ghosh, E.D. Kovetz, K.B. Body, A. Raccanelli (2025)  
**Phys. Rev. D 112, 063502 (2025), arXiv:2504.15348** - *Joint 21-cm and CMB Forecasts for Constraining Self-Interacting Massive Neutrinos*
- 12. G. Shmueli, S. Libanore**, E.D. Kovetz (2024)  
**Phys. Rev. D 111 063512 (2025), arXiv:2412.04071** - *Towards a multi-tracer neutrino mass measurement with line-intensity mapping*
- 11. J. Flitter, S. Libanore**, E.D. Kovetz (2024)  
**Phys. Rev. D 112 023537 (2025), arXiv:2412.04071** - *More careful treatment of matter density fluctuations in 21-cm simulations*
- 10. E. Vanzan, S. Libanore**, L. Valbulsa Dall'Armi, N. Bellomo, A. Raccanelli (2024)  
**JCAP 10 014 (2024), arXiv:2405.13871** - *Gravitational wave background from primordial black holes in globular clusters*
- 9. S. Libanore**, E.D. Kovetz (2024)  
**Phys. Rev. D 111 063521 (2024), arXiv:2404.01500** - *Upcoming searches for decaying dark matter with ULTRASAT ultraviolet maps*
- 8. S. Libanore**, E.D. Kovetz (2024)  
**A&A 687, A133 (2024), arXiv:2401.12285** - *Constraining  $z < 2$  ultraviolet emission with the upcoming ULTRASAT satellite*
- 7. S. Libanore**, J. Flitter, E.D. Kovetz, Z. Li, A. Dekel (2023)  
**MNRAS 532 1 (2024), arXiv:2310.03021** - *Effects of feedback-free starburst galaxies on the 21-cm signal and reionization history*
- 6. S. Libanore**, M. Liguori, A. Raccanelli (2023)  
**JCAP 08 055 (2023), arXiv:2306.03087** - *Signatures of primordial black holes in gravitational wave clustering*

FIND MY WORK  
ON INSPIRE HEP



5. M. Peron, A. Ravenni, **S. Libanore**, M. Liguori, M.C. Artale **(2023)**  
**MNRAS 530 1 (2024), arXiv:2305.18003** – *Clustering of binary black hole mergers: a detailed analysis of the EAGLE+MOBSE simulation*
4. T. Adi, **S. Libanore**, H.A. Cruz, E.D. Kovetz **(2023)**  
**JCAP 09 035 (2023), arXiv:2305.06440** – *Constraining Primordial Magnetic Fields with Line-Intensity Mapping*
3. **S. Libanore**, C. Unal, D. Sarkar, E.D. Kovetz **(2022)**  
**Phys. Rev. D 106 123512 (2022), arXiv:2208.01658** – *Unveiling cosmological information on small scales with line intensity mapping*
2. **S. Libanore**, M.C. Artale, M. Liguori, M. Mapelli et al. **(2021)**  
**JCAP 02 003 (2022), arXiv:2109.10857** – *Clustering of Gravitational Wave and Supernovae events: a multitracers analysis in Luminosity Distance Space*
1. **S. Libanore**, M.C. Artale, M. Liguori, M. Mapelli et al. **(2020)**  
**JCAP 02 035 (2021), arXiv:2007.06905** – *Gravitational Wave mergers as tracers of Large Scale Structures*

## TALKS & SEMINARS

- APG Seminar (SISSA, Trieste, May 2026)
- Astrophysics Seminar (Technion, May 2026)
- ARCO Seminar (Open University, January 2026)
- Dark Sector Meeting (Jet Propulsion Laboratory, November 2025)
- Cosmology Journal Club (Caltech, November 2025)
- Cosmology Group Meeting (University of Harvard, November 2025)
- Annual meeting of the Israeli Physical Society (Technion, July 2025)
- LIM 2025 (Annecy, June 2025)
- Cosmic Frontier Center Discussions (Austin, May 2025)
- SUBLIME meeting (Tokyo, March 2025)
- Astrophysics Seminar (Santander, February 2025)
- 21 cm Cosmology Workshop (Hangzhou, July 2024)
- LIM 2024 (Urbana Champaign, May 2024)
- Astrophysics Seminar (Weizmann Institute, April 2024)
- Cosmology in the Alps (Les Diablerets, March 2024)
- Astrophysics Seminar (Bar Ilan University, February 2024)
- Astrophysics Seminar (Scuola Normale Superiore di Pisa, November 2023)
- HI as a Cosmological probe (Nazareth, May 2023)
- Present and Future of LIM (Garching, April 2023)
- Annual meeting of the Israeli Physical Society (Tel Aviv, April 2023)
- Cosmology Journal Club (Padova, November 2022)
- Asiago Cosmology Group Retreat (Asiago, September 2022)
- ICTP Cosmology Summer School (Trieste, July 2022)
- Astrophysics Seminar (Ben Gurion University, March 2022)

- Gravitational Wave day @ Unipd (Padova, December 2021)
- AstroParticle Symposium 2021 (Saclay, November 2021)
- CosmoClassic (online, September 2021)
- SKAste (Tagliolo Monferrato, September 2021)
- First EuCAPT Annual Symposium (online, May 2021)
- IberiCOS 2021 (online, March 2021)
- University of the Western Cape Astro webinar (online, January 2021)
- CosmoMeetings at Universidade de São Paulo (online, July 2020)
- Cosmology Journal Club (Padova, June 2020)